

Advanced Cryptography

Elisabeth Oswald (module lead), Sam Collins (teaching assistant)

Module Overview, Administrative Stuff

THE CONTENT OF THIS MODULE

This module covers advanced topics in cryptography: this year with an emphasis on **secret sharing** and its application in the area of secure multiparty computation, and leakage resilient implementations.

If time permits, we will also look into threshold encryption and signatures, garbled circuits, as well as efficient zero-knowledge proofs via MPC in the head.

Introductory modules in cryptography tend to cover cryptographic techniques that are applied to “data at rest” and “data in transit”. **In this module we look at cryptographic techniques for “data in computation”.**

ORGANISATION

The **organisation** of this module is based on a **double session on Monday** (mostly lecture like) and a **so-called lab session on Tuesday** (somewhat interactive/group/pair work based).

I will, when it makes sense, use Tuesdays for making progress in terms of content initially, so that later on in the term, we can use more slots to help with coursework.

There is one lecturer only for this module, and one teaching assistant. Please use us wisely!

I can, if wanted, record the “lecture” parts of this course (live).

INTENDED FLOW OF TOPICS, FIRST HALF OF TERM

Week	Topic
W01	Admin, MPC examples and motivation, adversaries, provable security
W02	Secret sharing
W03	MPC based on secret sharing
W04	MPC based on secret sharing
W05	Embedded security: side channels and attacks, coursework due
W06	Recuperate and chill

In W01 we will use Tuesday to make progress with content (no lab). In W05 we might “swap” the lecture on Monday and lab on Tuesday (because the main lecturer is unavailable on Tuesday) (this depends on how well the Tuesday lab sessions are used).

INTENDED FLOW OF TOPICS, SECOND HALF OF TERM

Week	Topic
W07	Side channels: masking based on secret sharing
W08	Side channels: masking based on secret sharing
W09	Garbled circuits (lectures only)
W10	Mock exam, coursework due
W11	Talking through mock results

I intend to offer you the chance to sit a mock exam, so you get a feel for how I set exam questions: I will collect what you produce during the mock, and produce some feedback for week 11. The other slot in W10 is to offer last minute trouble shooting for coursework.

HOW DOES GRADING WORK?

The module is 50% exam and 50% coursework.

Coursework: Each labsheet (there will be two) has a fair number of formative and summative assessments. Formative assessments are exercises you can do by yourself, with a partner or in a group. They are only there to aid your understanding. **Summative** exercises are the ones that are graded. I will offer a fair number of different exercises (programming, paper and pencil, proofs), with different levels of difficulty and associated points.

You can freely choose which summative exercises you submit.

There will be **two submission deadlines** during the term: end of week 5 and end of week 10. Extensions for up to 5 days are possible. Waiving is possible with very good reasons.

You can get up to 25 points in the first submission and up to 25 points in the second submission.

COMMUNICATION AND HOW TO GET HELP

I do read/respond to emails ... but rather slowly.

“Office hour:” on Tuesdays directly after the lab session.

Please use the lab sessions to ask for help. The lab sessions are largely unstructured, and thus you are absolutely free to take staff aside and ask for help.

`m.e.oswald@bham.ac.uk`